

A comparison between Ethereum and Bitcoin

Features	Ethereum	Bitcoin
Block time	14 to 15 seconds	10 minutes
Mining	Generates new coins at a usually consistent rate.	Generates new coins at a rate that halves every four years.
Transaction features	Differ based on computational complexity, bandwidth use and storage needs.	Depend upon transaction size in bytes.
Units	Ethereum gas units are typically measured in Gwei.	Bitcoin transactions usually have fees specified in satoshis per byte.
Transaction fee	US\$ 0.156 (in June 2019).	US\$ 54 (in June 2019)

people earlier believed spread through the universe and allowed light to travel. Ethereum appeared in the market back in 2015, and is now one of the most reputable tokens.

As is the case with other cryptocurrencies, the validity of each Ether is provided by a blockchain, which results in blocks that are linked and secured using cryptography. It is an open, distributed ledger that records transactions between two parties efficiently and in a verifiable and permanent way. Ethereum doesn't rely upon Unspent Transaction Outputs (UTXOs). Here, current balances of all accounts and extra data are stored in a separate Merkle Patricia Tree.

Today, Ethereum is the second largest cryptocurrency in the world after Bitcoin. The latter's recent popularity has proven to be very positive in the crypto market. However, analysts now believe that Ethereum is likely to extend its upward surge towards US\$ 300 in the near future.

Challenges in Ethereum

Ether can be traded by regular currency brokers, on cryptocurrency exchanges as well as many online cryptocurrency wallets. Ethereum is a global, open source platform

for decentralised applications. On Ethereum, you can write code that controls digital values, which run exactly as programmed and are accessible anywhere in the world.

Until recently, building blockchain applications required a complex background in coding cryptography, mathematics as well as significant resources. But times have changed. By providing developers with the tools to build decentralised applications, Ethereum is making things easier.

Like Bitcoin, Ethereum is a distributed public blockchain network. In the Ethereum blockchain, instead of mining for Bitcoin, miners work to earn Ether, a type of crypto token that fuels the network.

The Ethereum blockchain has some extraordinary capabilities. One of them is that you can build smart contracts. It's a contract that self executes. It handles enforcement, the management, performance and payment.

Bitcoin and other cryptocurrencies exclusively operate as peer-to-peer digital currencies. Expanding the set of functions offered by Bitcoin is very complicated and time consuming. On the other hand, Ethereum makes use of a sort of Swiss Army Knife protocol. Ethereum's core innovation, the Ethereum Virtual Machine (EVM), is a Turing complete software that runs on

the Ethereum network. The EVM makes the process of creating blockchain applications much easier and more efficient than ever before. Ethereum potentially enables the development of thousands of different applications, all on one platform.

Any services that are centralised can be decentralised using Ethereum. Think of several services that exist across hundreds of different industries, intermediary services like title registries, voting systems, regularity compliance, etc. Ethereum can make such services easier and faster.

Benefits of Ethereum

So why do we need to move towards the Ethereum system? Here are some of the benefits that ought to push us towards making such a choice.

- **Immutability:** Third parties such as lawyers, courts or any such legal entities can't make any changes in the relevant data.
- **Corruption and tamper-proof:** Applications are based on a network formed and make censorship impossible.
- **Possibility of further growth:** Ethereum is not a self-locked system. This means that other goods and services can grow together with the Ethereum blockchain. It makes Ethereum grow continuously, and more and more people can start investing in it.
- **Security:** Ethereum does not fail and is secure as it uses cryptography. Applications here are protected against hacking attacks and fraudulent activities.
- **Possibility of improvement:** It is an open system, so everyone from outside can join this project and make it possible to improve this ecosystem.
- **Flexibility:** Flexibility is perhaps Ethereum's primary innovation. Ethereum allows users to create their own operations of any complexity they wish.
- **Zero downtime:** Applications never go down and can never be switched off.